Week 2 - Day 7

Canadian Housing Analysis - Final Project

A complete end-to-end data analysis project combining Pandas, NumPy, data cleaning, exploratory data analysis, Matplotlib, Seaborn, correlation analysis, visualization, and interpretation.

Stage	Focus	Status
1	Load & Inspect	Complete
2	Data Cleaning	Complete
3	Exploratory Data Analysis	Complete
4	Data Visualization	Complete
5	Final Insights & Conclusion	Complete

1. Stage 1 - Load & Inspect

Goal

Understand the dataset before changing anything.

• Loaded housing.csv with pandas.read_csv().

• Inspected the first rows, shape, columns, dtypes, info(), and describe().

• Initial dataset: 25 rows and 8 columns.

• Detected one missing Price, one missing Year_Built, and one duplicate row.

• Detected inconsistent categorical values such as london, VANCOUVER, condo, and CONDO.

Important inspection commands

```
df.head(10) df.shape df.columns.tolist() df.dtypes df.info() df.describe() df.isna().sum()
df.duplicated().sum()
```

2. Stage 2 - Data Cleaning

Categorical cleaning

```
df["City"] = df["City"].str.strip().str.title() df["Province"] =
df["Province"].str.strip().str.title() df["Property_Type"] =
df["Property_Type"].str.strip().str.title()
```

• Removed extra spaces and standardized capitalization.

• Removed the duplicate row, leaving 24 cleaned properties.

Missing values

• Price mean after duplicate removal: about $793,695.65.

• Price median: $690,000. Median was chosen as a simple fill because high-price properties pull the mean upward.

• Year_Built median: 2008. Missing Year_Built was filled and the column converted to integer.

Why median can help

For a right-skewed distribution, a few unusually high values can pull the mean upward. The median is more resistant to those extreme values. For a larger real dataset, a grouped imputation strategy using location and property type could be more representative.

Final validation

• 24 rows remained.

• No missing values remained.

• No duplicate rows remained.

• Year_Built became int64.

3. Stage 3 - Exploratory Data Analysis

Overall statistics

• Average property price: $789,375.

- Median property price: $690,000.
- Minimum price: $485,000; maximum price: $1,650,000.
- Average property area: about 1,605 sqft.
- Average bedrooms: 2.92; average bathrooms: 1.96.

Location findings

- Vancouver had the highest average city price: about $1.407M.
- British Columbia had the highest average provincial price: about $1.407M.
- The most expensive individual property was a detached Vancouver property priced at $1.65M.
- Calgary had the largest average property area among the cities in this sample.

Property type findings

- Detached: 10 properties; Condo: 9; Townhouse: 5.
- Detached properties had the highest average price ($886,000).
- Detached properties also had the largest average area (1,990 sqft).

Correlation

- Area_sqft vs Price: about 0.51 - moderate positive linear relationship.
- Year_Built vs Price: about 0.66 - strongest positive correlation with Price among the examined predictors.
- Bedrooms vs Area_sqft: about 0.97 - strongest relationship between two different variables in the matrix.

Key rule

Correlation describes association, not causation. A positive correlation does not prove that changing one variable directly causes the other to change.

4. Stage 4 - Data Visualization

Charts created

- Histogram + KDE: property price distribution.
- Bar plot: average property price by city.
- Scatter plot: Area_sqft vs Price, grouped by Property_Type.
- Correlation heatmap: numerical housing variables.
- Box plot: property prices by Property_Type.
- Independent province/area visualization experiment.

Histogram interpretation

- Most prices were concentrated roughly between $500,000 and $800,000.
- A few properties above $1M created a right-side tail.
- This right-skew helped explain why median was a reasonable simple imputation choice.

Box plot interpretation

• Detached properties had the highest median price.

• Detached properties also showed the widest price spread.

Scatter plot interpretation

• Price generally tended to increase as area increased, but the relationship was not perfect.

• Large properties were not automatically the most expensive; location, type, age, and other factors can matter.

• hue="Property_Type" added categorical grouping to each Area-Price observation.

5. Stage 5 - Final Insights & Conclusion

Final findings

• Property price in this sample varies with location and property characteristics.

• Vancouver, British Columbia appeared to be the most expensive market represented.

• Detached properties generally had larger areas and higher prices.

• Area_sqft and Year_Built both had positive relationships with Price; Year_Built was stronger in this dataset.

Dataset limitation

The cleaned dataset contains only 24 properties and only a few observations for each city and property type. Therefore, these results are useful for learning and demonstrating analysis techniques, but they should not be treated as representative of the entire Canadian housing market.

Complete workflow

Load Data -> Inspect Data -> Clean Data -> Analyze Data -> Visualize Data -> Interpret Results

6. Key Pandas / Visualization Commands

```
pd.read_csv() df.head() df.shape df.info() df.describe() df.isna().sum() df.duplicated()
df.drop_duplicates() df["col"].fillna() df.groupby() df.agg() df.sort_values() df.idxmax()
df.idxmin() df.corr() sns.histplot() sns.barplot() sns.scatterplot() sns.boxplot()
sns.heatmap() plt.figure() plt.title() plt.xlabel() plt.ylabel() plt.show()
```

7. Week 2 Completion

Day 1 NumPy - Complete
Day 2 Pandas - Complete
Day 3 Data Cleaning - Complete
Day 4 EDA - Complete
Day 5 Matplotlib - Complete
Day 6 Seaborn - Complete
Day 7 Canadian Housing Analysis - Complete

Week 2: 7/7 COMPLETE